

八年级数学参考答案

1. D 2. C 3. B 4. A 5. B 6. D 7. C 8. A

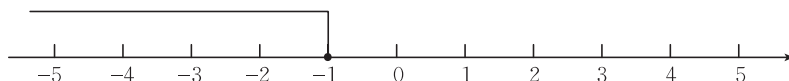
9. $(-1, 1)$ 10. $3x - 5 \leq 2$ 11. $x < -1$ 12. 2 13. 10 14. $0 < a \leq \frac{1}{3}$

15. 解: $3x - x \leq -2$,

$$2x \leq -2,$$

$$x \leq -1. \dots\dots\dots 3 \text{ 分}$$

该不等式的解集在数轴上表示如图所示. $\dots\dots\dots 5 \text{ 分}$



16. 解: 根据三角形的外角等于与它不相邻的两个内角的和, 得 $x + (x - 10) = x + 50$,
整理, 得 $2x - x = 50 + 10$,

$$\text{解得 } x = 60. \dots\dots\dots 5 \text{ 分}$$

17. 解: 解不等式①, 得 $x > -1$, $\dots\dots\dots 2 \text{ 分}$

解不等式②, 得 $x \leq 2$, $\dots\dots\dots 4 \text{ 分}$

$\therefore$ 不等式组的解集为 $-1 < x \leq 2$. $\dots\dots\dots 5 \text{ 分}$

18. 证明: 由题图可知, $AB^2 = 2^2 + 1^2 = 5$, $AC^2 = 4^2 + 2^2 = 20$, $BC^2 = 5^2 = 25$. $\dots\dots\dots 3 \text{ 分}$

$$\therefore AB^2 + AC^2 = 20 + 5 = 25 = BC^2,$$

$\therefore \triangle ABC$ 是直角三角形. $\dots\dots\dots 5 \text{ 分}$

19. 证明: $\because BD, CE$ 分别是 $\triangle ABC$ 的高,

$$\therefore \angle BEC = \angle CDB = 90^\circ.$$

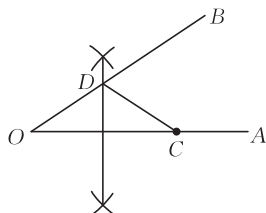
在 $\text{Rt}\triangle BEC$ 和 $\text{Rt}\triangle CDB$ 中,

$$\begin{cases} BC = CB, \\ BE = CD, \end{cases}$$

$\therefore \text{Rt}\triangle BEC \cong \text{Rt}\triangle CDB (\text{HL}), \dots\dots\dots 4 \text{ 分}$

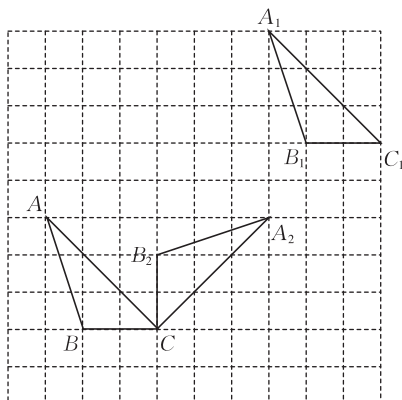
$\therefore EC = DB. \dots\dots\dots 5 \text{ 分}$

20. 解: 如图, $\angle CDB$ 即为所求. $\dots\dots\dots 5 \text{ 分}$



21. 解: (1) 如图, $\triangle A_1B_1C_1$ 即为所求. 3 分

(2) 如图, $\triangle A_2B_2C$ 即为所求. 6 分



22. 解:
$$\begin{cases} x+y=-5-m \text{ ①,} \\ x-y=3+3m \text{ ②,} \end{cases}$$

由①+②, 解得 $x=m-1$.

把 $x=m-1$ 代入①, 解得 $y=-4-2m$,

$\therefore$ 原方程组的解为 $\begin{cases} x=m-1, \\ y=-4-2m. \end{cases}$ 4 分

$\because$ 方程组的解满足 x 为非正数, y 为负数,

$$\therefore \begin{cases} m-1 \leq 0, \\ -4-2m < 0, \end{cases}$$

$\therefore -2 < m \leq 1$ 7 分

23. 解: (1) 证明: $\because FE \perp BC$,

$$\therefore \angle F + \angle C = 90^\circ.$$

$$\because \angle F = 30^\circ,$$

$$\therefore \angle C = 60^\circ. 2 \text{ 分}$$

$$\because AB = AC,$$

$\therefore \triangle ABC$ 是等边三角形. 3 分

(2) 由(1), 得 $\triangle ABC$ 是等边三角形,

$$\therefore \angle B = 60^\circ, AC = BC. 4 \text{ 分}$$

$$\because DE \perp BC,$$

$$\therefore \angle B + \angle BDE = 90^\circ,$$

$$\therefore \angle BDE = 30^\circ,$$

$$\therefore BE = \frac{1}{2} BD = 1,$$

$\therefore AC=BC=BE+EC=1+3=4$ 7 分

24. 解:(1)证明:由平移的性质,得 $AB \parallel EF, CD \parallel EP$,

$$\therefore \angle B = \angle EFP, \angle C = \angle EPF.$$

$$\because \angle B + \angle C = 90^\circ,$$

$$\therefore \angle EFP + \angle EPF = 90^\circ,$$

$$\therefore \angle FEP = 90^\circ,$$

$\therefore \triangle EFP$ 是直角三角形. 4 分

(2)由平移的性质,得 $EF=AB=8, EP=CD=6, AE=BF, ED=CP$.

由(1),得 $\triangle EFP$ 是直角三角形,

$$\therefore PF = \sqrt{EF^2 + EP^2} = \sqrt{8^2 + 6^2} = 10,$$

$$\therefore BF + PC = BC - PF = 15 - 10 = 5,$$

$$\therefore AD = AE + ED = BF + PC = 5. 8 分$$

25. 解:(1)设购进 B 型号服装的数量为 m 件,则购进 A 型号服装的数量为 $(2m+4)$ 件.

由题意,得 $2m+4 \leq 28$,

解得 $m \leq 12$.

答:B 型号服装最多可以购进 12 件. 3 分

(2)由题意,得 $18 \times (2m+4) + 30m \geq 765$,

解得 $m \geq 10.5$,

$$\therefore 10.5 \leq m \leq 12. 6 分$$

$\because m$ 为正整数,

$$\therefore m = 11 \text{ 或 } m = 12,$$

$\therefore$ 有两种方案:方案一,购进 A 型号服装 26 件,B 型号服装 11 件;

方案二,购进 A 型号服装 28 件,B 型号服装 12 件. 8 分

26. 解:(1)3. 2 分

(2)证明: $\because \angle BAC = 90^\circ, AB = AC$,

$$\therefore \angle ABD = \angle ACE = 45^\circ.$$

由旋转的性质可知, $\triangle AEC \cong \triangle AFB$,

$$\therefore \angle ABF = \angle ACE = 45^\circ, FB = CE, \angle FAB = \angle EAC, AF = AE,$$

$$\therefore \angle FBD = \angle ABF + \angle ABD = 90^\circ.$$

$$\because \angle BAC = 90^\circ, \angle DAE = 45^\circ,$$

$$\therefore \angle FAD = \angle FAB + \angle BAD = \angle EAC + \angle BAD = 90^\circ - 45^\circ = 45^\circ = \angle DAE.$$

$$\because AD = AD, AF = AE,$$

$$\therefore \triangle AFD \cong \triangle AED (\text{SAS}),$$

$$\therefore FD = DE.$$

在 $\text{Rt}\triangle FBD$ 中, 由勾股定理, 得 $BD^2 + BF^2 = DF^2$,

$$\text{即 } BD^2 + CE^2 = DE^2. \dots\dots\dots 7 \text{ 分}$$

(3) 能. 如图, 连接 AD , 将 $\triangle ABD$ 绕点 A 逆时针旋转至 $\triangle ACE$, 过点 A 作 $AF \perp DC$ 于点 F .

由旋转性质, 得 $\angle ABD = \angle ACE = 120^\circ$, $BD = CE = 60$ 米, $AD = AE$, $S_{\triangle ABD} = S_{\triangle ACE}$.

$$\because \angle ACD = 60^\circ,$$

$$\therefore \angle ACD + \angle ACE = 180^\circ,$$

$\therefore$ 点 E, C, D 共线.

$$\because AD = AE, AF \perp DE,$$

$$\therefore DF = EF.$$

在 $\text{Rt}\triangle AFC$ 中, $\angle ACD = 60^\circ$,

$$\therefore \angle CAF = 30^\circ,$$

$$\therefore FC = \frac{1}{2} AC = 50 \text{ (米)}.$$

$$\text{由勾股定理得 } AF = \sqrt{AC^2 - FC^2} = \sqrt{100^2 - 50^2} = 50\sqrt{3} \text{ (米)}. \dots\dots\dots 9 \text{ 分}$$

$$\because EC = 60 \text{ 米},$$

$$\therefore EF = EC + FC = 60 + 50 = 110 \text{ (米)},$$

$$\therefore DE = 2EF = 220 \text{ (米)}.$$

$$\because S_{\text{四边形}ABDC} = S_{\triangle ACD} + S_{\triangle ABD}, S_{\triangle ABD} = S_{\triangle ACE},$$

$$\therefore S_{\text{四边形}ABDC} = S_{\triangle ACD} + S_{\triangle ACE} = S_{\triangle ADE}.$$

$$S_{\triangle ADE} = \frac{1}{2} DE \cdot AF = \frac{1}{2} \times 220 \times 50\sqrt{3} = 5\,500\sqrt{3} \text{ (平方米)},$$

$$\text{即四边形 } ABDC \text{ 的面积为 } 5\,500\sqrt{3} \text{ 平方米}. \dots\dots\dots 12 \text{ 分}$$

